Ex: No: 3 Ex: 3 - Experiments on CISCO Packet Tracer (simulation tool).

Aim:

To study the packet tracer tool installation and user interface Overview.

c) To understand environment of CISCO Packet tracer to design simple network.

Introduction:

1) It allows you to model complex systems without the need for dedicated equipment.

2) It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.

3) It is available for both the linux and windows desktop environments.

4) Protocols in Packet tracker are coded to work and behave in the same way as they would on real hardware.

Installing Packet tracer.

To download Packet tracer, go to https://www.netacad.com and log in with your cisco networking academy credentials; then, click on the packet tracer graphic and download the package appropriate for your operating system.

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User interface overview:

The layout of packet tracer is divided into several components. The components of the packet tracer interface are as follows; match the numbering with explanations.

1) Menu bar - This is a common menu found in all software applications; it is used to open, save, print, change preferences, and so on.

2) Main toolbar - This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo and redo and on the right-hand side is an icon for entering network information for the current network.

3) Logical/physical workspace tab - These tabs allow you to toggle between the logical and physical work areas.

4) Workspace - This is the area where topologies are created and simulations are displayed.

5) Common tools bar - This toolbar provides controls for manipulating topologies.

6) Real time / Simulation tabs - These tabs are used to toggle between the real and simulation modes.

7) Network component box - This component contains all of the network and end devices available with packet tracer and is further divided into two areas; Device type selection box and Device-specific selection box.

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c) User-created packet box - users can create highly customized packets to test their topology from this area, and the results are displayed as a list.

d) Analyse the behaviour of network devices using Cisco packet tracer simulator:

i) From the network component box, click and drag and drop the below components.

a) 4 generic PCs and One hub.

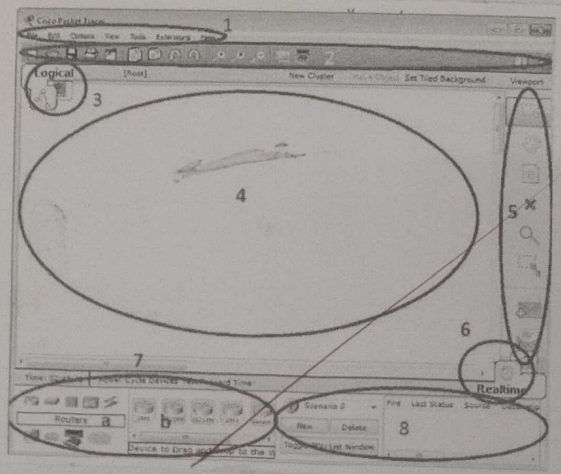
b) 4 generic PCs and one switch.

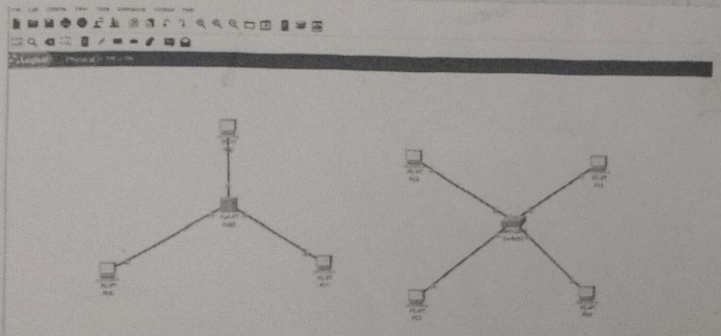
ii) Click on connections,

a) Click on copper straight-through cable.

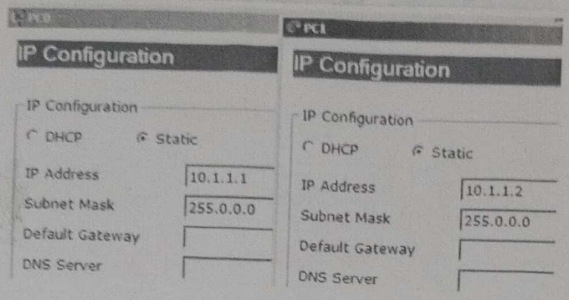
b) Select one of the PC and connect it to hub using the cable. The link LED should glow in green, indicating that the link is up. similarly connect remaining 3 PCs to the hub.

iii) Similarly connect 4 PCs to the switch using copper straight-through cable





iv) Click on the Pc's connected to hub, go to the Desktop tab, click on IP configuration and enter an IP address and subnet mask. Here the default gateway and DNS server information is not needed as there are only two end devices in the network.



v) Observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation.

vi) Repeat step 3 to step 5 for the Pc's connected to the switch.

vii) Observe how hub and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and hub.

Student observation:

a) From your observation write down the behaviour of switch and hub in terms of forwarding the packets received by them.

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⇒ Hub:

* A hub broadcasts the received packet to all ports connected to it, except the port from which the packet was received.

Switch:

* A switch checks the MAC address of the destination device and forwards the packet only to the specific port where the device is connected.

Device	Packet forwarding Behaviour.
Hub	Broadcast packets to all connected ports.
Switch	forwards packets only to the destination port.

b) Find out the network topology implemented in your college and draw and label that topology in your observation book.

⇒ Most modern college computer networks use a star topology, often extended into a hierarchical/ extended star topology.

Labels to include:

* Internet.
* Access switch
* Network cables.
* Core switch.

Observation:

The college network follows an extended star topology, where multiple computers are connected to access switches and the access switches are connected to a central switch.

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